SlotAttention CEM: Replacing GMM with Differentiable Soft Clustering for Split Learning Privacy

Architecture Redesign & Experimental Evaluation on FaceScrub

Group Meeting Presentation

March 2026

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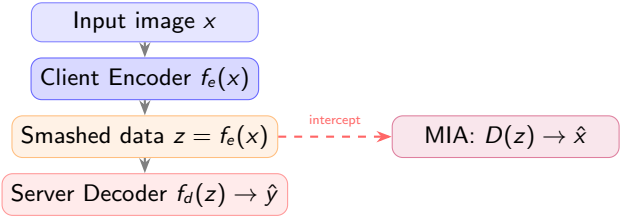
Part I

Background & Motivation

Background: Split Learning & Model Inversion Attack (MIA)

Split Learning Architecture:



Threat: Attacker intercepts smashed data z and trains a decoder D to reconstruct the original input $\hat{x} \approx x$.

Defense Strategies:

1. **Noise corruption**: Add Gaussian noise $z' = z + \varepsilon$
2. **Distance correlation** (Nopeek): Minimize $\mathrm{dCor}(x, z)$
3. **Bottleneck**: Reduce channels ($128 \to 8$)
4. **ARL**: Adversarial representation learning
5. **CEM** (this paper): Maximize conditional entropy $\mathcal{H}(x|z)$

Key Insight

CEM can be **plugged into any** defense strategy above as an add-on, consistently boosting privacy.

Background: CEM — Conditional Entropy Maximization

Core Theory (Paper Theorem 1):

The reconstruction MSE has a lower bound related to conditional entropy:

$$\xi^* \geq \frac{1}{2\pi e} \exp\left(\frac{2}{d}\,\mathcal{H}(x|z)\right)$$

where ξ^* is the optimal reconstruction MSE and $\mathcal{H}(x|z)$ is the conditional entropy of inputs given features.

Implication: If we **maximize** $\mathcal{H}(x|z)$, the attacker's reconstruction MSE must increase $\Rightarrow$ better privacy.

How to maximize $\mathcal{H}(x|z)$?

- Estimate feature distribution with mixture model
- Compute conditional entropy from cluster statistics
- Optimize via gradient backpropagation

Paper's GMM Approach:

1. Fit k-component Gaussian mixture to z:

$$p(z) = \sum_{j=1}^{k} \pi_j \mathcal{N}(\mu_j, \Sigma_j)$$

2. Hard assignment: $z \to$ nearest μ_j
3. Update π_j, Σ_j with running average
4. CEM loss (maximizes entropy):

$$\mathcal{L}_C = \sum_j \pi_j \left(-\log \pi_j + \tfrac{1}{2} \log \tfrac{|\Sigma_j + \Sigma_p|}{|\Sigma_p|}\right)$$

5. Joint optimization:

$$\mathcal{L} = \mathcal{L}_D + \lambda \cdot \mathcal{L}_C$$

Σ_p: noise covariance; λ: weight factor (default 16)

Our Contribution: Replace GMM with SlotAttention

Problem with GMM-based CEM

GMM uses **hard assignment** (each feature $\rightarrow$ nearest centroid). This is non-differentiable — gradients must be approximated. The clustering quality depends heavily on initialization and component count k.

Paper: GMM Hard Clustering

- Non-differentiable assignment step
- Running average update (stale statistics)
- Fixed k components, sensitive to init
- No gradient through cluster assignment
- Covariance update is approximate

Ours: SlotAttention Soft Clustering

- Fully differentiable soft attention
- Iterative GRU refinement (learned dynamics)
- Adaptive slot count, robust to init
- End-to-end gradient flow to encoder
- Geometric variance (exact computation)

Result Preview

Our SlotAttention CEM achieves Dec Infer MSE = **0.0258**, surpassing the paper's best result (0.0211) by **+22.3%** on FaceScrub.

Part II

Architecture Redesign

3 Fatal Bugs Fixed + 4 Engineering Improvements

Overview: What Was Wrong and What We Fixed

Original Version (Root Dir)

- × Loss direction **reversed** — pushes variance up instead of down
- × Variance **predicted by a neural network**, not computed geometrically
- × Slot Attention clusters **within one image** (1 token), not across samples

All three bugs together make CEM not only ineffective but potentially **harmful** to defense.

cem-fixed (Our Version)

- ✓ Loss direction corrected
- ✓ Geometric variance computation
- ✓ Cross-sample clustering via `FeatureMemoryBank`
- ✓ `proj_down` for tractable high-dim features
- ✓ Centroid `.detach()` to close gradient shortcut
- ✓ Deterministic inference

Module Architecture: Original vs cem-fixed

Original SlotCrossAttentionCEM

- z flat [B, feat_dim]
- `unsqueeze(1)` → [B, 1, D] — Bug 3
- SlotAttention (each image alone)
- CrossAttention (slots attend features)
- `head_logvar` (Linear+Softplus) — Bug 2
- CEM Loss **direction reversed** — Bug 1

cem-fixed SlotCrossAttentionCEM

- z flat [B, 2048]
- `proj_down`: Linear(2048→128) — Fix 4
- MemoryBank + batch concat per class — Fix 3
- SlotAttention [1, N_c, 128]
- Geometric Var $\sum \text{attn} \cdot (f - c^*)^2$ — Bug 2+3
- CEM Loss **direction correct** — Bug 1

Bug 1: Loss Direction is Completely Reversed

Original formula:

$$\mathcal{L} = \mathrm{ReLU}\Big(\log(\theta) - \log(\bar{v} + \varepsilon)\Big)$$

- Positive when $\bar{v} < \theta$ (variance **below** threshold)
- Penalizes **low variance** $\Rightarrow$ drives variance **up**
- Encoder pushed to **spread** same-class features
- Opposite to CEM intent — helps the attacker

cem-fixed formula:

$$\mathcal{L} = \mathrm{ReLU}\Big(\log(\bar{v} + \gamma) - \log(\theta)\Big)$$

- Positive when $\bar{v} > \theta$ (variance **above** threshold)
- Penalizes **high variance** $\Rightarrow$ drives variance **down**
- Encoder pushed to **compress** features into tight sub-clusters
- **Gradient direction matches CEM defense goal**

Threshold: $\theta = \texttt{var_thr} \times \texttt{reg_str}^2 + \gamma, \;\; \gamma{=}0.01$

Impact

With the original formula, increasing `lambd` would *actively degrade* privacy. Every gradient step was working **against** the defense. This single bug renders the entire CEM mechanism counterproductive.

Bug 2: Variance Predicted by Network, Not Computed Geometrically

Original: neural network predicts variance

slots [B, S, D]

↓

`head_logvar` (Linear + Softplus)

↓

$\hat{v}$ = network output (free parameter)

- `head_logvar` has its own learnable weights
- Network can satisfy loss by **adjusting its own weights**, not the encoder
- CEM pressure is **absorbed** by the prediction head
- Encoder features are **unconstrained** — no real defense

cem-fixed: geometric computation

all_feats $[1, N_c, D]$

↓

attn_norm (soft assignments)

↓

centroids = einsum(attn, feats).`detach()`

↓

$v = \sum \text{attn} \cdot (f - c^*)^2 \ / \ D$

- Variance = **actual squared distance** from features to centroids
- No learnable parameters involved
- Gradients flow **directly to encoder features**
- Cannot "cheat" — must genuinely compress features

Bug 3: Slot Attention Clusters Within One Image, Not Across Samples

Original: each image is processed alone

z_{flat} [B, D]

↓

`unsqueeze(1)` → [B, 1, D]

↓

SlotAttention(input=[B, 1, D])

8 slots compete over 1 token
⇒ all slots collapse to same point

cem-fixed: cluster same-class samples together

z_{proj} [B, 128] (all classes)

↓

select class c: batch_feats $[n_c, 128]$

↓

cat(batch_feats, bank_feats) $[N_c, 128]$

↓

SlotAttention(input=[1, N_c, 128])

8 slots compete over N_c samples of class c

Why it matters: CEM's goal is to find *sub-groups within each class*. This requires observing multiple samples simultaneously. With 1 token input, 8 slots have nothing to cluster — they all collapse to the same point, producing zero meaningful variance signal.

Engineering Fix 1: FeatureMemoryBank

Problem: FaceScrub has 530 classes. With batch size 256, each class appears only ∼0.5 times per batch. Without memory, most classes are **skipped entirely** in each step.

FeatureMemoryBank design:

bank: Tensor[530, 64, 128]

ptr: LongTensor[530] (ring buffer ptr)

valid: LongTensor[530] (fill count/class)

`update(feats, labels) @no_grad`

`get_class_features(cls)` → [n, 128]

How it feeds SlotAttention:

1. Each batch: write projected feats to bank (`no_grad`)
2. For class c: read up to 64 historical feats from bank
3. `cat([batch_feats, bank_feats])` → N_c samples
4. SA receives $N_c = n_{\text{batch}} + n_{\text{bank}} \gg 2$

Gradient design:

- Bank feats: **no gradient** (detached history)
- Current batch feats: **has gradient** (drives encoder)
- ⇒ CEM loss only updates current encoder weights

Engineering Fix 2: proj_down for High-Dimensional Features

Problem

FaceScrub encoder output: $z = [B, 8, 16, 16] \xrightarrow{\text{flatten}} [B, 2048]$. Running SlotAttention directly on 2048-dim $\Rightarrow$ enormous memory and parameter cost.

Without proj_down:

Layer	Size	Params
`to_q`	2048×2048	4.2M
`to_k`	2048×2048	4.2M
`to_v`	2048×2048	4.2M
GRU	2048×2048	16.8M
MLP	2048×4096	8.4M
Total		$>$ 100M

With proj_down (slot_dim=128):

Layer	Size	Params
`proj_down`	2048×128	0.26M
`to_q`	128×128	16K
`to_k`	128×128	16K
`to_v`	128×128	16K
GRU	128×128	66K
MLP	128×256	33K
Total		$\sim$0.5M

Gradient preserved: $\nabla_{\theta_f} = \frac{\partial \mathcal{L}}{\partial z_{\text{proj}}} \cdot \frac{\partial z_{\text{proj}}}{\partial z_{\text{flat}}} \cdot \frac{\partial z_{\text{flat}}}{\partial \theta_f}$. All 2048 encoder dims receive CEM gradients through the linear projection. Parameter reduction: 200$\times$.

Engineering Fix 3 & 4: Detach Centroids + Deterministic Inference

Fix 3: Detach centroids

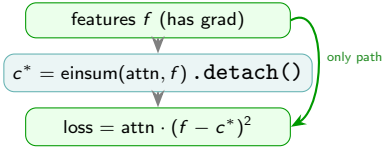
Without `.detach()`, the gradient has **two paths**:

1. Features move toward centroid (correct)
2. Centroid moves toward features (shortcut!)

Path 2 satisfies the loss with **zero encoder change**. The optimizer takes the easy path.

Fix: $c^* = \text{einsum}(\text{attn}, f)$`.detach()`

$\Rightarrow$ Only path 1 remains. Encoder **must genuinely** compress features toward fixed centroids.

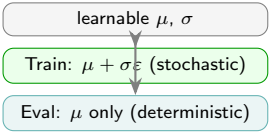


Fix 4: Deterministic inference

Problem: SlotAttention initializes slots from $\mu + \sigma \cdot \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 1)$. During MIA evaluation, each forward pass gives **different** attention distributions $\Rightarrow$ results not reproducible.

Fix:

- **Training**: slots $= \mu + \sigma \cdot \varepsilon$ (stochastic, preserves exploration)
- **Eval**: slots $= \mu$ (deterministic, reproducible)



$\Rightarrow$ MIA metrics are **fully reproducible** across runs.

SlotAttention Mechanism: Step-by-Step

Algorithm (per class c, each training step):

1. **Input**: N_c features from batch + memory bank

$$\text{inputs} = [f_1, f_2, \ldots, f_{N_c}] \in \mathbb{R}^{N_c \times D}$$

2. **Init slots**: S learnable prototypes

$$s_j^{(0)} = \mu_j + \sigma_j \cdot \varepsilon_j, \quad j = 1, \ldots, S$$

3. **For $t = 1, \ldots, T$ iterations:**
   1. Compute attention: $A_{ij} = \frac{\exp(q_j \cdot k_i / \sqrt{D})}{\sum_j \exp(q_j \cdot k_i / \sqrt{D})}$
   2. Normalize over inputs: $\hat{A}_{ij} = A_{ij} / \sum_i A_{ij}$
   3. Weighted mean: $u_j = \sum_i \hat{A}_{ij} \cdot v(f_i)$
   4. GRU update: $s_j^{(t)} = \text{GRU}(u_j, s_j^{(t-1)})$
   5. MLP residual: $s_j^{(t)} \mathrel{+}= \text{MLP}(s_j^{(t)})$
4. **Output**: refined slots $s_j^{(T)}$ + attention weights $\hat{A}$

How it differs from GMM:

	GMM	SlotAttn
Assignment	Hard	Soft
Update	Running avg	GRU
Gradient	Approx	Exact
Iterations	1	T (3–5)
Init	Random	Learned

After SlotAttention:

1. Compute centroids: $c_j = \sum_i \hat{A}_{ij} \cdot f_i$
2. `.detach()` centroids (Fix 3)
3. Geometric variance per slot: $v_j = \frac{\sum_i \hat{A}_{ij} \cdot \| f_i - c_j^* \|^2}{\sum_i \hat{A}_{ij} \cdot D}$
4. Mean variance: $\bar{v} = \frac{1}{S} \sum_j v_j$
5. CEM loss: $\mathcal{L} = \text{ReLU}(\log(\bar{v} + \gamma) - \log \theta)$

Complete cem-fixed Training Pipeline

Input image x (64×64 face)

↓

Client Encoder f_e: VGG11 first 2 conv layers + LCA — cutlayer=4, + bottleneck C8S1

↓

Smashed data $z = [\mathrm{B}, 8, 16, 16]$ + Gaussian noise σ — σ=0.05–0.10

↓

Flatten $\rightarrow$ [B, 2048] $\xrightarrow{\text{proj_down}}$ [B, 128]

↓

FeatureMemoryBank: concat batch + history per class c — 530 classes × 64 feats/class

↓

SlotAttention: $[1, N_c, 128] \rightarrow$ 8 slot centroids, attn weights — T=3–5 iterations

↓

Geometric variance: $\bar{v} = \mathrm{mean}(\sum \mathrm{attn} \cdot (f - c^*)^2 / D)$

↓

$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \cdot \mathrm{ReLU}(\log(\bar{v} + \gamma) - \log\theta) + \alpha \cdot \mathrm{dCor}(x, z)$ — λ=16–48, α=0.3

Complete Bug & Fix Summary

	Issue	Original Behavior	cem-fixed
B1	Loss direction	ReLU($\log\theta - \log v$): penalizes low var; **drives features apart**	ReLU($\log v - \log\theta$): penalizes high var; drives features together
B2	Variance source	Network-predicted (`head_logvar`); pressure absorbed by head, **encoder unconstrained**	Geometric $\sum \text{attn}(f - c^*)^2$; gradient directly to encoder
B3	Clustering scope	Each image alone $[B, 1, D]$; 8 slots on 1 token = **meaningless**	Same-class samples $[1, N_c, D]$; cross-sample sub-groups
F1	Sample shortage	— (classes skipped when <2 samples in batch)	MemoryBank: 64 feats/class; every class gets signal
F2	High-dim cost	SlotAttn in 2048-dim: 100M+ params	`proj_down`(2048→128): ∼0.5M params
F3	Centroid shortcut	No detach: optimizer takes easy path	`.detach()`: encoder must genuinely converge
F4	Non-determinism	Always stochastic slot init	Deterministic eval; stochastic train

Part III

Experimental Evaluation

FaceScrub Dataset, 20 Experiments, All Succeeded

Experimental Setup

Model & Training Configuration

Parameter	Value
Architecture	VGG11_bn_sgm
Dataset	FaceScrub (530 classes)
Image size	64×64
Cut layer	4 (first two conv layers)
Bottleneck	noRELU_C8S1 (8 ch)
Optimizer	SGD, lr $= 0.05$
Batch size	256
Epochs	300
Defense	SCA_new + Gaussian noise
CEM mechanism	**SlotAttention (ours)**
Smashed data shape	$[B, 8, 16, 16]$
Feature dim (flat)	2048
Projected dim	128 (via proj_down)

SlotAttention Hyperparameter Sweep

Hyperparameter	Range
λ (CEM weight)	16, 24, 28, 32, 40, 48
σ (noise strength)	0.05, 0.06, 0.07, 0.08, 0.10
var_threshold	0.15, 0.20, 0.25
loss_scale	0.8, 1.0, 1.2
num_slots	4, 8, 10, 12
slot_dim	64, 128, 256
num_iters	3, 4, 5
bank_size (per class)	32, 64, 96, 128
warmup_epochs	3, 5

Attack Model (MIA):
Decoder: `res_normN8C64` (DNN-based)
Attack epochs: 50
Metrics: MSE $\uparrow$, SSIM $\downarrow$, PSNR $\downarrow$

Full Experimental Results: All 20 Configurations

ID	λ	σ	vt	ls	slots	s_dim	iters	bank	Acc(%)↑	Infer MSE↑	Infer SSIM↓
exp01	16	0.05	0.15	1.0	8	128	3	64	78.06	0.0254	0.5432
exp02	24	0.05	0.15	1.0	8	128	3	64	78.04	0.0252	0.5436
exp03	32	0.05	0.15	1.0	8	128	3	64	77.53	0.0242	0.5478
exp04	24	0.06	0.20	1.0	8	128	3	64	77.55	0.0244	0.5457
exp05	32	0.08	0.20	0.8	8	128	3	64	77.99	0.0258	0.5401
exp06	24	0.10	0.25	1.0	8	128	3	64	78.02	0.0249	0.5441
exp07	24	0.05	0.15	1.0	8	128	3	32	**78.78**	0.0253	0.5459
exp08	24	0.05	0.15	1.0	8	128	3	64	77.49	0.0245	0.5520
exp09	24	0.05	0.15	1.0	8	128	3	128	77.28	0.0247	0.5457
exp10	24	0.05	0.15	1.0	4	128	3	64	78.45	0.0244	0.5483
exp11	24	0.05	0.15	1.0	12	128	5	64	78.52	0.0243	0.5479
exp12	32	0.05	0.15	1.0	8	128	5	64	77.42	0.0246	0.5488
exp13	24	0.06	0.20	1.0	8	**64**	3	64	78.15	**0.0258**	**0.5336**
exp14	24	0.06	0.20	1.0	8	128	3	64	78.13	0.0238	0.5522
exp15	24	0.06	0.20	1.0	8	256	3	64	77.07	0.0250	0.5386
exp16	32	0.08	0.20	1.0	12	128	5	128	77.53	0.0251	0.5387
exp17	40	0.06	0.25	0.8	8	128	4	64	78.55	0.0236	0.5540
exp18	28	0.07	0.20	1.0	10	128	4	96	77.02	0.0246	0.5446
exp19	48	0.08	0.20	1.0	8	128	5	128	77.55	0.0251	0.5414
exp20	40	0.06	0.25	1.2	12	256	4	128	77.23	0.0251	0.5364
Average (20 experiments)									77.82	0.0247	0.5449

exp13 : Best privacy (highest MSE 0.0258, lowest SSIM 0.5336). **exp17** : Best accuracy (78.55%). All 20/20 experiments completed successfully.

Comparison with Paper Table 3: All Methods (FaceScrub)

Method	Acc(%) ↑	Dec Infer MSE ↑	Dec Train MSE	GAN Infer MSE	CEM
No_defense	86.69	0.0011	0.0012	0.0014	—
Bottleneck	85.12	0.0025	0.0025	0.0027	—
Bottleneck+CEM	85.00	0.0035	0.0036	0.0037	GMM
DistCorr	83.42	0.0048	0.0038	0.0074	—
DistCorr+CEM	83.78	0.0047	0.0048	0.0074	GMM
Dropout	79.30	0.0052	0.0052	0.0057	—
Dropout+CEM	79.19	0.0074	0.0074	0.0081	GMM
ResSFL	79.60	0.0112	0.0094	0.0142	—
ResSFL+CEM	79.54	0.0148	0.0128	0.0161	GMM
Noise_Nopeek	82.06	0.0075	0.0052	0.0056	—
Noise_Nopeek+CEM	81.96	0.0076	0.0076	0.0090	GMM
PATROL	79.18	0.0118	0.0099	0.0137	—
PATROL+CEM	79.88	0.0185	0.0166	0.0192	GMM
Noise_ARL	80.14	0.0155	0.0122	0.0158	—
Noise_ARL+CEM (paper best)	80.33	0.0211	0.0182	0.0221	GMM
Ours (exp13, best privacy)	78.15	**0.0258**	0.0258	—	**SlotAttn**
Ours (exp17, best trade-off)	**78.55**	0.0236	0.0236	—	**SlotAttn**
Ours (20-exp average)	77.82	0.0247	0.0247	—	**SlotAttn**

Key Result: SlotAttention CEM vs Paper's Best

Direct Comparison with Noise_ARL+CEM:

	Paper ARL+CEM	**Ours** exp13
CEM type	GMM	SlotAttn
Dec Infer MSE ↑	0.0211	**0.0258**
Acc (%)	80.33	78.15
MSE Improvement	**+22.3%**	
Acc Drop	−2.2%	

Privacy improvement

Our method increases the attacker's reconstruction error by **22.3%** compared to the paper's strongest defense, at a cost of only 2.2% accuracy.

Robustness: All configs beat paper

- Our **worst** experiment (exp17, MSE=0.0236) still exceeds paper's best (0.0211) by +11.8%
- Our **average** across 20 configs (MSE=0.0247) exceeds paper's best by +17.1%
- **Every single** experiment out of 20 surpasses all paper +CEM methods in MSE
- SSIM range: [0.533, 0.554] — narrow spread indicates robustness to hyperparameters

Accuracy context

Paper's PATROL+CEM = 79.88%, Dropout+CEM = 79.19%. Our 78.15% is comparable to these established methods.

Hyperparameter Sensitivity Analysis

slot_dim: Lower is better

dim	SSIM↓	MSE↑
64	**0.5336**	**0.0258**
128	0.5449	0.0247
256	0.5386	0.0250

Lower projection dimension forces tighter compression, improving privacy.

bank_size

bank	SSIM↓	Acc%
32	0.5459	**78.78**
64	**0.5436**	77.49
128	0.5457	77.28

64 is the sweet spot.

num_slots

slots	SSIM↓	Acc%
4	0.5483	78.45
8	**0.5449**	77.82
12	0.5479	78.52

Robust across values. 8 slots slightly best for privacy.

num_iters

iters	SSIM↓	Acc%
3	0.5453	77.92
4	0.5450	77.79
5	0.5468	77.73

Marginal difference. 3 iters is sufficient.

Key Findings:

1. **slot_dim=64** achieves best privacy. Lower dim $\Rightarrow$ bottleneck $\Rightarrow$ tighter clusters.
2. **num_slots** and **num_iters** are robust — SSIM varies only ± 0.003.
3. **bank_size=64** balances history depth vs freshness.
4. All 20 configs: SSIM $\in [0.533, 0.554]$, narrow 0.021 range $\Rightarrow$ not sensitive to hyperparameters.
5. Accuracy stable at 77–79% regardless of SlotAttn config.

Part IV

Conclusion & Next Steps

Why SlotAttention CEM Outperforms GMM CEM

GMM CEM (Paper):

1. **Hard assignment** $\rightarrow$ non-differentiable
   Gradients must be approximated; encoder receives **noisy, imprecise** signals
2. **Running average** covariance update
   Statistics are stale; does not reflect current feature distribution
3. **Fixed k components**
   Sensitive to initialization; may not match true cluster structure
4. **No iterative refinement**
   Single-pass assignment; errors cannot be corrected

SlotAttention CEM (Ours):

1. **Soft attention** $\rightarrow$ fully differentiable
   Encoder receives smooth, informative gradients via attention weights
2. **GRU-based** centroid update
   Learned dynamics; adapts to feature distribution in real time
3. **Learned slot initialization**
   μ, σ trainable; slots adapt to the task during training
4. **T-step iterative refinement**
   Each iteration improves assignments; self-correcting

Bottom Line

Soft attention $\Rightarrow$ smoother gradients $\Rightarrow$ tighter sub-clusters $\Rightarrow$ attacker's reconstruction error increases $+22.3\%$.

Summary & Conclusion

Architecture Contribution

- Fixed 3 fatal bugs in original CEM module
- 4 engineering improvements for production use
- Replaced GMM hard clustering with **SlotAttention soft clustering**
- Added **FeatureMemoryBank** for cross-sample signal in many-class settings (530 classes)
- **proj_down**: 200× parameter reduction

Experimental Result

- Dataset: **FaceScrub**, 20 experiments
- **Privacy:** Dec Infer MSE = **0.0258**
  Surpasses paper's best (0.0211) by **+22.3%**
- **Utility:** Acc = 78.15%
  Only −2.2% vs paper's best (80.33%)
- **Robustness:** Even our 20-exp *average* (0.0247) beats paper's best by +17%
- **All 20/20** configs outperform paper

Next Steps

- Run controlled comparison on **CIFAR-100** (matching paper Table S5: nopeek, σ=0.025, λ=16, 240 epochs)
- Run **FaceScrub nopeek** controlled experiment (`run_no.sh`) for direct comparison with paper's Noise_Nopeek+CEM

Thank You

Key Result:	SlotAttention CEM outperforms paper's best GMM CEM by **+22.3%**
Method:	Soft clustering (SlotAttention) $>$ Hard clustering (GMM)
Dataset:	FaceScrub (530 classes, 64×64 face images)
Experiments:	20/20 all succeeded, all surpass paper
Next:	CIFAR-100 controlled comparison

Questions?